The FREUID Challenge 2026

Technical Report — Template-Aligned Field-Crop MIL Forensics

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Abstract

We address binary detection of forged identity documents across five template-locked document types (four driving licences, one ID card; Latin and Arabic scripts) under the DET-based FREUID score (lower is better). Our key observation is that all documents of a type are pixel-aligned to a common template, while whole-image downsampling (e.g. to 384^2) renders document text $\sim$10 px tall, making character-level tampering physically invisible to a whole-image classifier. Our system therefore scores documents *per field*: for each type we derive up to 13 field bounding boxes (photo, name, date fields, document number, MRZ, plus three static zones) from per-pixel variance templates over bona-fide images, cut the crops at native resolution, and score each crop with a shared EfficientNet-B3 fed channel-concatenated RGB and SRM noise-residual streams. The document fraud score is the maximum over crop scores (multiple-instance learning), with horizontal-flip test-time augmentation. Trained only on the official training set (ImageNet-initialised backbone), this single model improves the public leaderboard from 0.15185 (our best whole-image baseline) to **0.03905**, a 3.9× reduction. Code (MIT), frozen weights, and a no-network Docker container reproduce the submission end-to-end.

Competition: The FREUID Challenge 2026 (IJCAI-ECAI 2026)
Kaggle team: jgshin22
Code: https://github.com/sdlvkqmv/freuid-challenge-2026

1 Introduction

The challenge asks for a continuous fraud score per document image; the metric, FREUID $= 1 - \mathrm{HM}(1 - \mathrm{AuDET}, 1 - \mathrm{APCER@1\%BPCER})$, is computed from the full DET curve, so ranking quality and the strict 1%-BPCER operating point matter while monotonic rescaling does not. The threat model mixes physical manipulation, generative-AI edits, and print-and-capture recapture [1]. The central practical difficulty is a train/test domain gap: near-perfect in-domain validation ($\sim 10^{-4}$ FREUID) coexists with public-leaderboard scores two to three orders of magnitude worse, so local CV cannot rank candidate models.

2 Method

2.1 Overview

Pipeline: (1) assign each test image a document type by 1024-bit difference-hash (dhash) nearest-neighbour matching against all 69,352 training images (documents are template-locked, so this is near-perfect); (2) resize to the type's canonical resolution; (3) cut $K \leq 13$ predefined field crops and resize each to 320^2; (4) score every crop with a shared CNN; (5) document score $= \max_k \sigma(\mathrm{logit}_k)$, averaged over two horizontal-flip views.

2.2 Template-aligned field boxes

Per type we compute per-pixel mean/standard-deviation templates over aligned bona-fide images. High-variance pixels localise per-document ink; horizontal dilation and vertical closing merge glyphs into field bands, and connected components give candidate boxes. We keep the portrait photo, up to nine field bands (name, dates, document number, MRZ, hologram areas), and three fixed static-background zones, padded by 3% (`assets/field_boxes_v2.json`).

2.3 Model architecture

A single EfficientNet-B3 (timm `tf_efficientnet_b3`, ImageNet initialisation, ∼11M parameters) with 6 input channels: RGB plus a fixed 3-channel SRM high-pass noise residual (Fridrich–Kodovský kernels). The scorer is shared across crops; a document is a bag of K=13 crop instances (missing boxes padded by repetition) and the document logit is the max over per-crop logits — one tampered field suffices to make a document fraudulent.

2.4 Training objective and procedure

BCE-with-logits on the document logit with pos_weight = 1.363 (class balance); AdamW, LR 3×10^{-4} with one-epoch warmup and cosine decay, weight decay 0.05, 6 epochs, batch 3 documents (39 crops), AMP, gradient clipping 1.0, seed 42. Augmentation: horizontal flip (p=0.5) and ColorJitter(0.1); no random-resized-crop (it would break template alignment). Best-validation checkpoint (epoch 4) selected.

2.5 Score calibration

None. Submitted scores are raw sigmoid probabilities; the metric is invariant to monotonic transforms. A per-type rank normalisation was tried and *regressed* the leaderboard (0.571), so no post-processing is applied.

3 Data

3.1 FREUID training set

All 69,352 official training images; labels from `train_labels.csv`. `is_digital` is degenerate (69,332 of 69,352 True) and unused. Stratified 5-fold split by (`label`×`type`); fold 0 held out for validation.

3.2 External data and pre-trained models

No external data beyond FREUID and publicly licensed ImageNet-pretrained `timm` weights (Apache-2.0), used only for backbone initialisation at training time.

3.3 Validation strategy

In-domain validation saturates (10^{-4}–10^{-7} FREUID) and does not track the leaderboard; model selection therefore used public-leaderboard submissions (30+ documented attempts, `docs/` in the repository). Private test labels were never used; private-row predictions were produced once, after the July 13 private release, with the frozen checkpoint.

4 Inference

Exact command (also the Docker entrypoint): `python submission/prepare_submission.py` with `/data` (flat image directory) and `/submissions` mounted. Steps: dhash type assignment → canonical resize → 13 field crops at 320^2 → shared scorer → hflip-TTA max-aggregation. Batch 6 documents (78 crops); one 10 GB GPU processes the 7,821 public images in a few minutes (CPU ∼20–30× slower). Output: `submission.csv` with `id,label`.

5 Results

Table 1: Main results. Lower FREUID is better.

Model	Local val FREUID	Public LB FREUID
Whole-image EffB3+SRM+recapture-aug (best of 20+ variants)	$\sim 10^{-4}$	0.15185
Field-crop MIL v1 (K=10, 384px)	$\sim 10^{-6}$	0.08023
Field-crop MIL v2 (K=13, 320px, hflip, static zones)	1.2×10^{-7}	**0.03905**

Notable negative results (all public-LB verified): DCT input stream, frozen CLIP/DINOv2 probes, test-time BN adaptation, hard-negative mining, blind multi-crop aggregation, synthetic recapture-realism augmentation, face-consistency fusion, diverse-backbone and reseed ensembles of the whole-image model, per-type rank normalisation. The single change that broke the whole-image plateau was *where the model looks* (native-resolution field crops with max aggregation), not the loss, stream set, or backbone.

6 Reproducibility

- **Repository:** https://github.com/sdlvkqmv/freuid-challenge-2026 (MIT); the frozen commit SHA is stated in our Kaggle discussion reply.
- **Docker:** `docker build -t freuid-repro:local -f submission/Dockerfile .` from the repo root; weights (43 MB), field boxes, and dhash references are baked into the image (`assets/`, SHA-256 in `assets/CHECKSUMS.txt`).
- **Dependencies:** Python 3.11, PyTorch 2.4.1, timm 1.0.27 (pinned in `submission/requirements.txt`); CUDA 12.1 optional.
- **Runtime:** full public-test inference in minutes on one 10 GB GPU; training ∼6 h on the same GPU.
- **Randomness:** seed 42 everywhere; cuDNN autotune enabled (training non-bitwise-deterministic; the frozen checkpoint removes this concern at inference).
- **Network:** inference runs under `-network none`; no runtime downloads.

7 Team and contributions

Solo entry: Gyucheol Shin (modelling, experimentation, engineering, writing), assisted by AI pair-programming tools for implementation and experiment management.

References

[1] FREUID Challenge Organizers. The freuid challenge 2026: Forged and manipulated identity document detection. https://www.kaggle.com/competitions/the-freuid-challenge-2026-ijcai-ecai, 2026. IJCAI-ECAI 2026 competition.

A Hyperparameters

Hyperparameter	Value
Crop size / count	320^2 / K=13 per document
Canonical sizes	BENIN,MAURITIUS 1585×1000; EGYPT 1387×875; GUINEA 1584×1000; MOZAMBIQUE 1000
Batch size	3 documents (39 crops)
Learning rate	3×10^{-4}, 1-epoch warmup, cosine decay
Weight decay	0.05
Epochs	6 (best-val = epoch 4)
Loss	BCE, pos_weight 1.363
Augmentation	hflip 0.5, ColorJitter 0.1
TTA	horizontal flip (2 views)
Seed	42